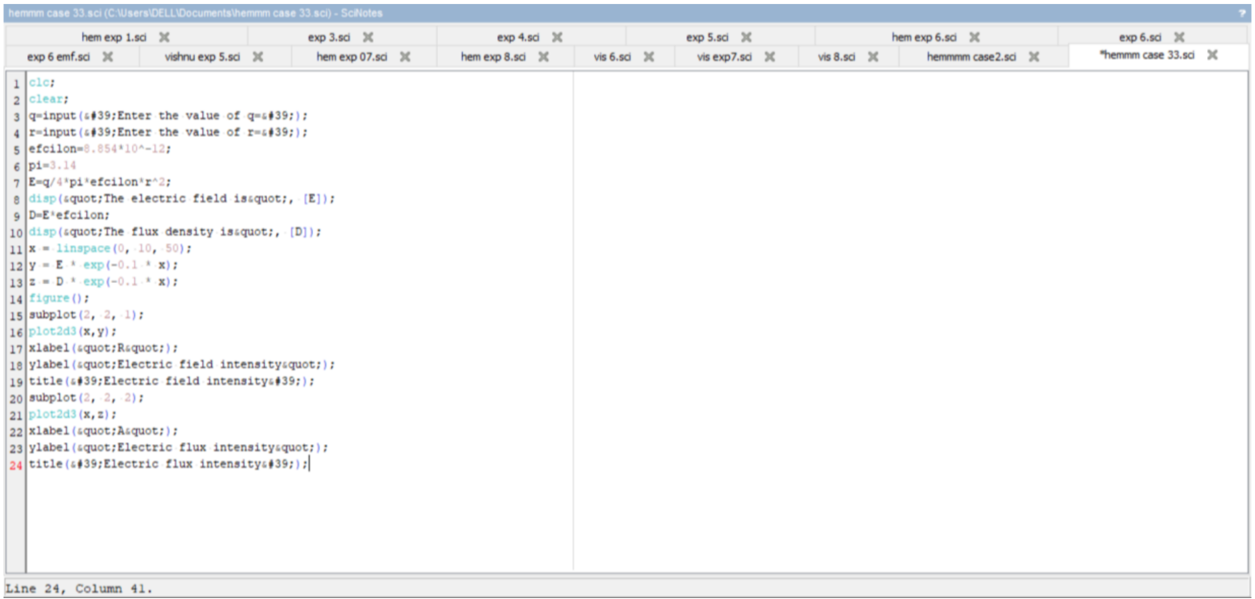
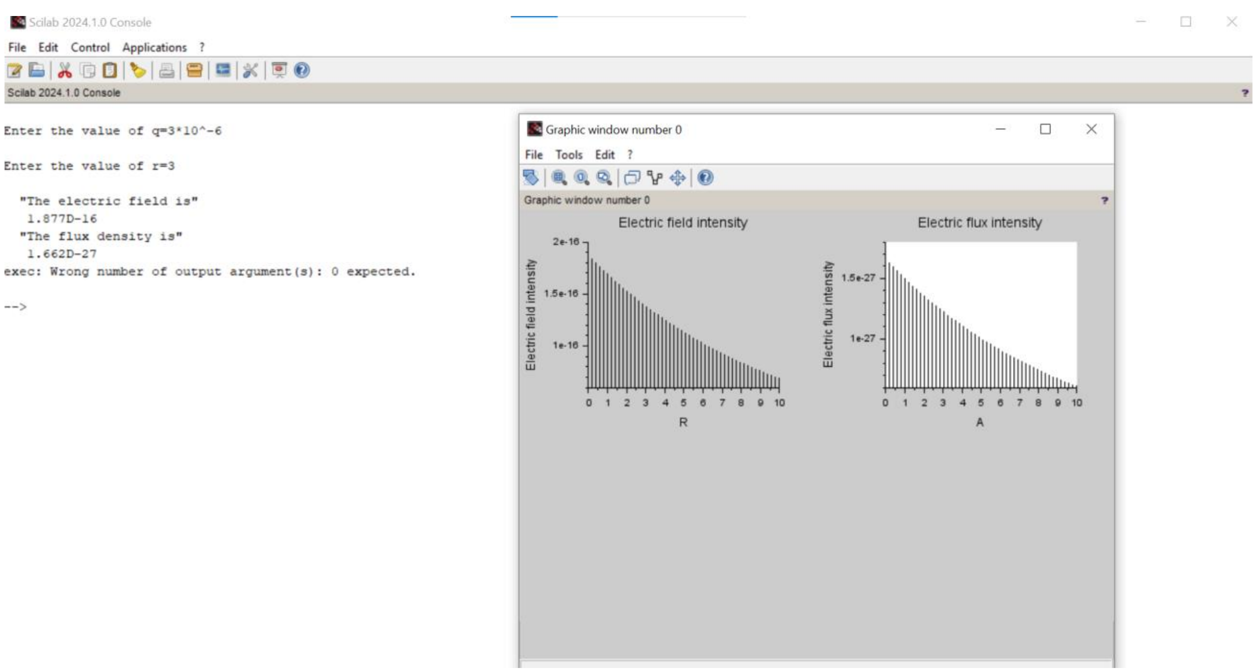
EXP 2: Determination and plotting of Electric field and Electric flux density in sphere full of charges with varying radius.





Ex No:- 2

Determination and plotting of electric field flux density is charges with varying radius

$Q = 3 \times 10^{-6}$

$r = 3\,cm$.

Electric field $E = \frac{Q}{4\pi \vartheta_E}$

flux density $u = E = 2$

The electric field is

~~1.1980e+09~~ 1.8770-16

The flux density is

~~1.3240e-20~~ 1.6620-27

Electric field

$$E = \frac{Q}{4\pi\varepsilon_0 r^2}$$

$$E = \frac{3\times10^{-6}}{4\pi(8.854\times10^{-12})(3\times10^{-2})^2}$$

$$E = \frac{3\times10^{-6}}{4\pi(8.854\times10^{-12})(9\times10^{-4})}$$

$$\boxed{E = 1.8470\times10^{-9}}$$

flux density

$$D = \varepsilon_0 E$$

$$D = (8.854\times10^{-12})(3.00\times10^{7})$$

$$\boxed{D = \cancel{1.6620\times10^{-20}}}$$

$$\boxed{D = 1.6620.}$$